Temporal Olympic Medal Prediction with Performance Momentum

UE24CS352A Machine Learning mini-project, Team 53. Ryona Aditree Ghosh (PES1UG24AM233) and Rishit V N S (PES1UG24AM225). Code: github.com/ryonahh/ML_MiniProject_53

1. Problem statement

The reference paper (Dobkowski, *2020 Summer Olympics Predictions Using Machine Learning*) predicts each country's medal count by combining World Bank indicators with the previous Games' medals, using a two-stage classifier-then-regressor model. We ask a narrower question: **using only information available before a Games, does explicitly modelling a country's performance momentum improve medal forecasts?** Last Games' count is already a strong forecast, so any gain must come from how a country's trajectory is represented.

2. Dataset

We use Kaggle's *120 years of Olympic history* (athlete_events.csv, 271,116 athlete-event rows; noc_regions.csv). We keep Summer Games only and drop the unofficial 1906 Games. The raw file repeats a team medal once per athlete, so medals are de-duplicated per (NOC, Games, event, medal), which reproduces the official tables (2016: USA 121, China 70, UK 67). NOC codes are mapped to regions so historical teams form one history (FRG/GDR to Germany, URS/EUN to Russia). The result is 2,770 Country × Games rows; the target is *Total_Medals*. The real Tokyo 2020 medal table is used only to score our 2020 forecast.

3. Approach

Leakage-free features. Every feature for Games *t* comes from the country's earlier appearances (group-wise shift): previous medals and golds; previous athletes, sports and events; medal growth and athlete growth as log-differences between the last two Games; a three-Games rolling average; and medal efficiency (medals per athlete).

Novelty: Performance Momentum. We test momentum in two ways. (i) A *momentum score*: the mean of the z-scored medal growth, athlete growth and efficiency, with the scaler fitted inside the sklearn pipeline on training Games only. (ii) A *momentum target* (our proposal): the Random Forest learns $\log(1+\text{medals}_t) - \log(1+\text{medals}_{t-1})$ and the forecast is $(1+\text{medals}_{t-1})\cdot e^{\text{pred}} - 1$. A forest can only output averages of targets it has seen, so learning counts pulls large nations toward the middle; learning change lets one leaf serve a 3-medal and a 90-medal nation on the same trajectory.

Controlled comparison. Six models share one Random Forest (300 trees, min_samples_leaf=10, max_features=0.6), chosen once by mean validation MAE on 2012 across all models, so only features and targets differ: Baseline (previous medals and golds), Historical Features, Historical + Momentum Score, the paper's Two-Stage design, Momentum Target, and Momentum Target + Score. We train on 1964–2012 and test on Rio 2016 (five seeds), backtest every Games from 1996 to 2016 with an expanding window, run a paired bootstrap on the 2016 errors, and forecast Tokyo 2020 from data up to 2016.

4. Implementation

A Python package (`src/olympic_momentum`): `data.py` (download, clean, aggregate), `features.py` (lagged features and the `PerformanceMomentum` transformer), `models.py` (a `MedalForecaster` that wraps the pipeline and the target encoding, plus the two-stage model), `experiment.py` (tuning, test, backtest, bootstrap, forecast) and `reporting.py`. One command, `python scripts/run_pipeline.py`, regenerates every table, figure and the website data. Twelve pytest tests check team-medal counting, that changing a future result never alters past features, and that every model fits and predicts non-negative counts. A demo notebook and an interactive website (docs/index.html) present the results.

5. Results

On the held-out Rio 2016 Games the Momentum Target model reaches **MAE 1.298**, RMSE 3.37 and R^2 0.933: **19.2%** lower MAE than the baseline and **11.9%** lower than the same nine features predicting the count. The paired bootstrap gives a 95% interval of 0.02 to 0.34 fewer medals of error per country (99% of resamples favour our model). In the backtest a momentum-target model has the lowest error in all six Games, 1996–2016.

Model	MAE	RMSE	R^2	Top-10 MAE	Backtest MAE
Baseline	1.606	3.85	0.913	12.26	1.629
Historical Features	1.474	3.76	0.916	11.91	1.599
Historical + Momentum Score	1.501	3.81	0.914	12.26	1.618
Paper Two-Stage	1.490	3.92	0.909	12.29	1.558
Momentum Target	**1.298**	**3.37**	**0.933**	**10.14**	**1.374**
Momentum Target + Score	1.296	3.39	0.932	10.16	1.389

Table 1. Rio 2016 held-out test (mean of five seeds) and average MAE over the six-Games backtest. Lower is better except R^2.

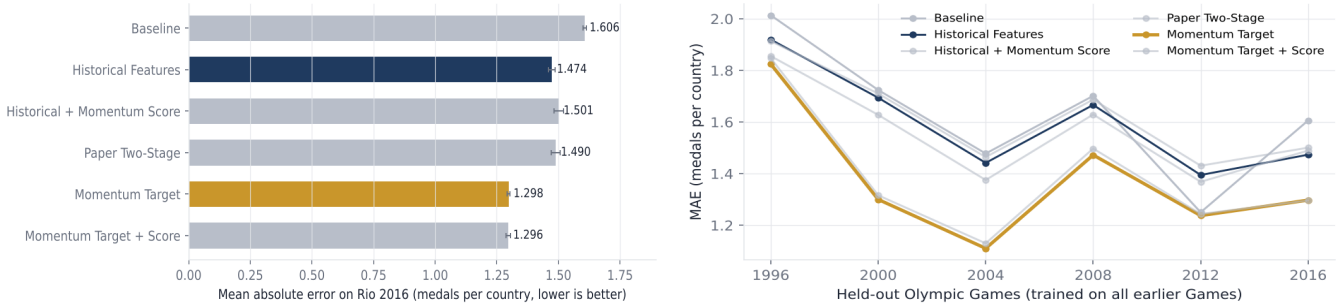


Figure 1. Left: Rio 2016 MAE per model (gold = proposed, navy = historical features). Right: MAE in each backtest Games.

Two results did not support our initial expectations, and we report them as found. First, adding the momentum *score* as a feature does not help (MAE 1.501 vs 1.474): the forest already has its three components. Second, on the genuinely unseen Tokyo 2020 Games every model beyond the baseline finishes level (MAE 1.55–1.57; Momentum Target 1.557, baseline 1.851). Tokyo was postponed by COVID and hosted by Japan (58 medals, up from 41), shocks that Olympic history alone cannot anticipate. Permutation importance ranks previous medals first and medal efficiency second.

6. Conclusions

How momentum is represented matters more than adding it. Turning it into the learning target (predicting each country's change rather than its count) gives a consistent and statistically supported improvement over the count model and the reference paper's two-stage design on identical data, while the composite score alone adds nothing. The approach is leakage-free and reproducible from one public dataset. Its limits show on Tokyo 2020, which motivates future work: host-nation flags, socioeconomic covariates as in the reference paper, count-aware gradient boosting (Poisson/Tweedie), and event-level models.